

THE OPERATION — the minimal generating set of BST (one object, one act, every observable a reading)

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THE OPERATION

One geometry. One act. Everything else is a reading.

The one sentence (for a bright high-schooler)

There is one shape — a specific 5-dimensional geometry called D_{IV}^5 — and one thing that happens on it: a commitment. To commit is to write a single yes/no record you can't take back, and to wrap it in the smallest self-checking code so it can't be quietly corrupted. Write the bit, check the bit. Quantum mechanics, spacetime, and the particles are three different ways of *reading* that one act. That is the whole theory: a shape, and an act that writes to it.

The one act (the operation), stated once

A commitment is one operation with two stages:

- WRITE — an *irreversible, binary* record deposited on the discrete boundary (the Shilov boundary). “This happened; it cannot be un-happened.” (The arrow of time = the one-sidedness of the write.)
- ENCODE — the continuum answers with a *confined 3-structure* that lets a corruption be *detected and located* (an erasure code, minimum distance 2 — detection/erasure, not error-correction). The record is thus self-validating — it carries its own damage report — *not* “unforgeable” (a public code forges trivially; the tamper-resistance is the irreversibility, not the code).

One act, two stages, neither omissible: drop the encode $\rightarrow$ the record can be silently corrupted; drop the write $\rightarrow$ there is nothing to encode. [Tier: this decomposition = the Peirce split of the commitment idempotent; verified. The “encode = d=2 detection” corrected the earlier “error-correction” reading — Elie 5303.]

As one operator (linear algebra on D_{IV}^5)

The operation is a single graded operator on the spin factor:

$C = P_{\text{record}} \oplus P_{\text{encode}}$ — a rank-1 projection onto the arrow line (the binary write, eigenvalues $\{0,1\}$) $\oplus$ the projection onto V_{12} (the confined 3-encode).

This is the geometry's *own* decomposition (the Peirce split of a primitive idempotent), not a story laid over it. [Tier: Derived — Grace's graded-operator form, Keeper-verified.]

The three readings (Internal-B: everything is a face of the one operator)

Reading	What it is	Tier
Quantum mechanics	The commit ontology derives all 10 Dirac–von Neumann axioms, zero posits (measurement = commitment; the Born statistics are the ENSEMBLE record; the arrow = the write’s one-sidedness).	DONE (10/10) — the standout.
Spacetime	The write distinguishes one positive/time direction (SO(2) circle) → Lorentzian signature; the boundary carries the 3 spatial directions; time is the common <i>append-axis</i> threading every commit. The observed (3,1) is the matter-induced descent (T2565).	ADVANCED — type forced (below); the (3,1) descent is matter-induced, not geometry-forced.
Standard Model	Fermions = the SVD of one FK overlap on D_IV ⁵ ; masses = singular values, mixing = vectors, CP = phases. Addresses forced, many masses derived.	STRONG, one honest holdout (below).

Is the geometry itself forced? (Internal-A — the forced-object result)

Commitment forces the *type* of the geometry; the *dimension* is the honest residual. The chain, all theorem/verified:

binary commitment $\boxtimes$ rank 2 $\boxtimes$ isotropic complement $\boxtimes$ spin factor $\boxtimes$ type-IV (Lorentzian) geometry.

Honest four-posit accounting (nothing hidden): - P0 — the commitment structure is a Euclidean Jordan algebra (the arena; not derived). - P1 — a commitment is one *binary* distinction (Casey’s yes/no; grounded by the Born rule as the resolving act, but Born is downstream so it *grounds* not *forces*). $\boxtimes$ rank 2 $\boxtimes$ Lorentzian type-IV, and by the same isotropy that gives spacetime no preferred frame (one principle, two jobs). - P2a — exclusivity (“genuinely type-IV, no second name”) — *unmotivated*; does the dimensional work. - P2b — minimality — but minimality is degenerate in every lane (it gives rank 1, the smallest code [3,1,3], and $n=3$, never the target). So the dimension $n_C = 5$ is NOT commitment-forced — it is one gated equation ($g = N_C^2$ –rank, a rank-2 coincidence in a dense integer ring; possible-numerology). [Tier: type forced mod P0+P1; dimension open. Grace’s boundary theorem: a geometry cannot force its own defining parameters — they are the definition-level boundary.]

External forcing achieved: ONE BST integer (rank = 2), from Shannon (binary alphabet = the outcome count), non-circularly — the first time any BST integer has been forced from *outside* the geometry. ($N_C=3$ is a candidate — the coding $d_{\min}=3 \rightarrow$ color-3 map is owed and obstructed [antisymmetry is $d=2$, not $d=3$].)

The information face — “Forcing + Evidence = 2 + 3”

The two externally-forced integers *are* the two halves of a trustworthy commitment: rank 2 = the FORCING (the binary decision) + $N_C = 3$ = the EVIDENCE (the validation/parity). BST’s *epistemology* (how the theory is defended: forcing + evidence) mirrors its *ontology* (what a commitment is: decide + validate). [Tier: framing + a sharpening of the honest forcing floor; names the two forced integers, does not reach the other three.]

The Standard-Model reading — where it stands (honest)

Derived: the generation addresses $\nu=\{5/2,3/2,0\}$ (forced 3 ways, T2517); down + charged-lepton mass ratios (forward closed forms, T2003); Cabibbo $V_{us}=1/\sqrt{20} + m_s/m_d=20$ (blind, T2529/30). The one honest holdout — Koide’s amplitude — is IDENTIFIED (open in physics; forcing ruled BREAK, K1221): the relation $Q=2/3$ is exact

to 0.001%, but the generations are a *dying ladder* (odd cohomology $h^1 \rightarrow h^3 \rightarrow h^5$, $h^7=0$), not a cyclic *ring*, so no symmetry forces the 45°/equal-norm balance; it is a numerical balance the field has not derived either. Falsifier (Koide 1981, not BST): $m_\tau = 1776.969 \text{ MeV}$, 0.91σ , dies if $\sigma(m_\tau) \rightarrow 0.03$. [The corrected geometric statement: “ $A^2=2$ from equal-norm,” never “ $A^2=\text{rank}$ ” (a rank-2 coincidence).]

What “complete” means, and where we are

- Postulates: one geometry, five integers, zero dimensionless free parameters — shorter than GR.
- QM: DONE (10/10).
- Spacetime/GR: ADVANCED (type forced; descent matter-induced; publish owed).
- SM parameters: STRONG with one characterized holdout (Koide’s amplitude, open in physics).
- Predictions: the Six Absences; DESI $w_a > 0$; θ_{23} octant; Bell $1/2^{\wedge} N_c$; $\Sigma m_\nu(m_1=0)$; $0\nu\beta\beta$ 1–4 meV.

The credibility signal is the *shape*, not a “100%”: a crank claims every cell complete; BST tiers each one precisely, keeps its corrections in the text, and names its one open holdout as physics’ open problem.

The honest floor (the durable legacy)

BST is forced down to: a Jordan arena + “a commitment is one binary distinction” (an axiom, the cheapest possible — “one act is one act”) + minimality. Everything downstream — QM, the induced GR framework, ~150 constants — follows. The act of commitment writes the world; the geometry is the ledger it writes on; and every observable is a way of reading that ledger.

v0.1, 2026-08-16. Synthesis of the forced-object/operation arc (K1593–K1624). Tiers current as of session end; corrections (self-validating≠unforgeable, encode=d=2, $A^2=2 \neq A^2=\text{rank}$, N_c candidate, $n_C=5$ open) are in the text by design. Nothing pushed. CP existence-only.